Assignment-3

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Course: Computer Networks Lab

Course Code: (ENCS304)

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Experiment Objectives:

Create a network simulation to demonstrate RIP (Routing Information Protocol) and OSPF (Open Shortest Path First) using Cisco Packet Tracer. Compare the performance, convergence, and efficiency of both routing protocols by simulating data transmission across multiple network .

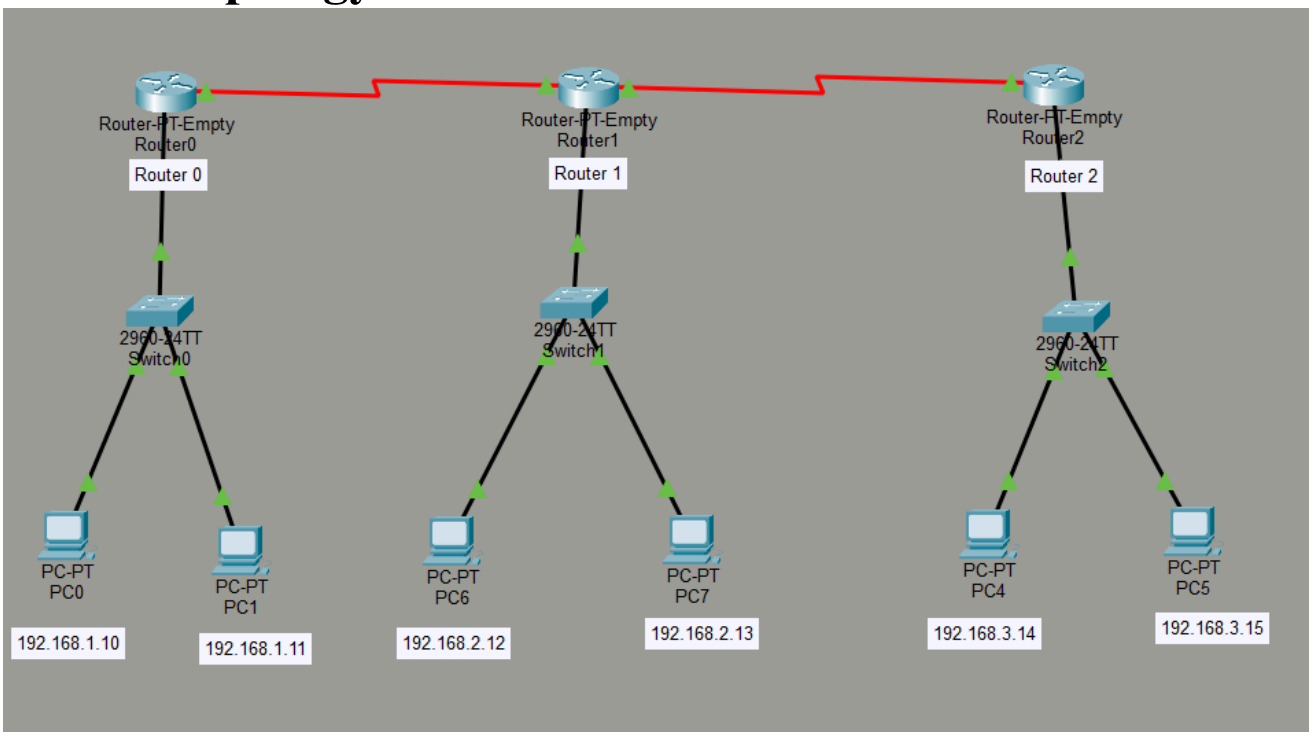
Learning Outcomes

- Apply routing concepts using RIP and OSPF
- Design and implement routing simulations in Cisco Packet Tracer
- Analyze routing tables and path selection
- Compare performance and convergence of routing protocols
- Evaluate routing behavior using simulation mode

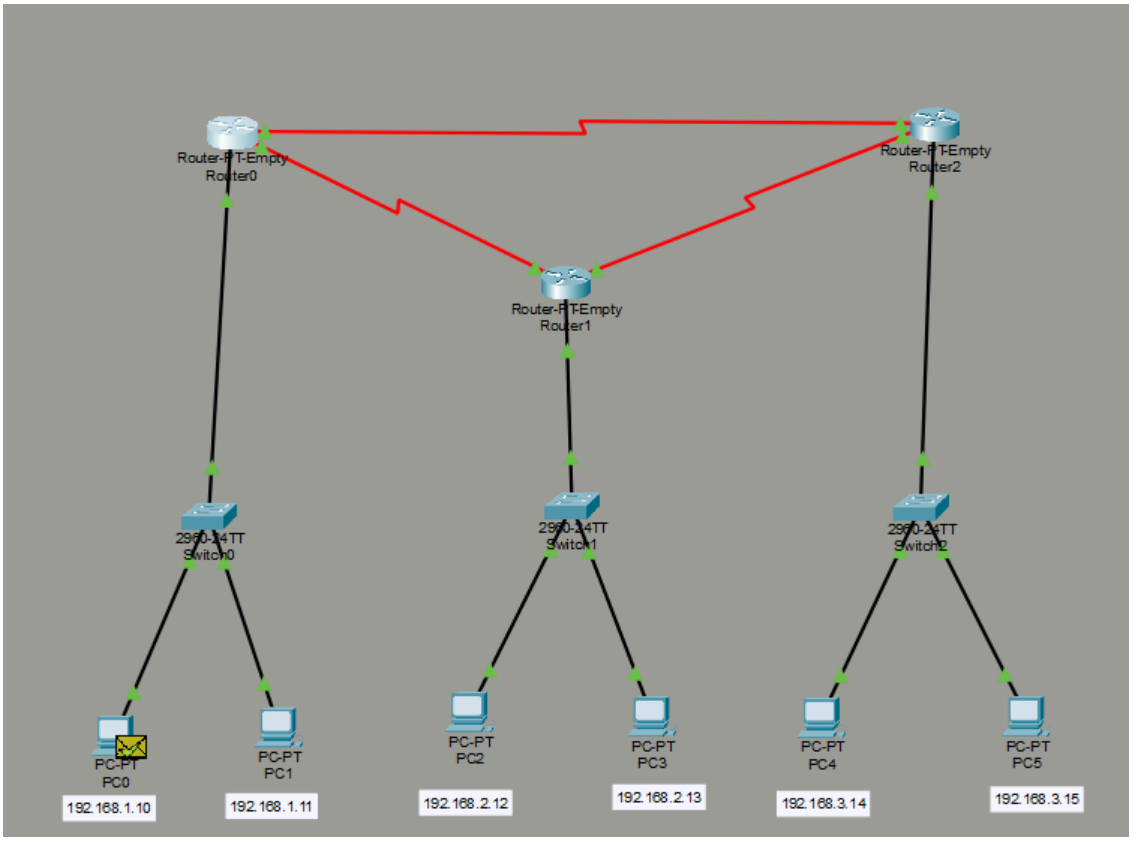
Logical structure should include:

- Network Devices: Routers, Switches, PCs
- Network Connections: Copper cables
- Routing Protocols: RIP and OSPF
- Data Flow Simulation: Packet transmission between networks

Linear Topology :



Triangle Topology :



Routing Tables:

1. Triangle Topology

- **RIP Routing Table**

Copper Straight-Through Connections

From	To	Port/Interface
PC0	S1	fa0/1
PC1	S1	fa0/2
S1	R0	fa0/24 → fa2/0
PC2	S2	fa0/1
PC3	S2	fa0/2
S2	R1	fa0/24 → fa2/0
PC4	S3	fa0/1
PC5	S3	fa0/2
S3	R2	fa0/24 → fa2/0

Serial DTE Connections

From	To	Port/Interface
R0	R1	se0/0 ↔ se1/0
R1	R2	se0/0 ↔ se1/0
R0	R2	se1/0 ↔ se0/0

- **OSPF Routing Table**

Copper Straight-Through Connections

From	To	Port/Interface
PC0	S1	fa0/1
PC1	S1	fa0/2
S1	R0	fa0/24 → fa2/0
PC2	S2	fa0/1
PC3	S2	fa0/2
S2	R1	fa0/24 → fa2/0
PC4	S3	fa0/1
PC5	S3	fa0/2
S3	R2	fa0/24 → fa2/0

Serial DTE Connections

From	To	Port/Interface
R0	R1	se0/0 ↔ se1/0
R1	R2	se0/0 ↔ se1/0
R0	R2	se1/0 ↔ se0/0

📘 2. Linear Topology

🔹 RIP Routing Table

Copper Straight-Through Connections

From	To	Port/Interface
PC0	S1	fa0/1
PC1	S1	fa0/2
S1	R0	fa0/24 → fa2/0
PC2	S2	fa0/1
PC3	S2	fa0/2
S2	R1	fa0/24 → fa2/0
PC4	S3	fa0/1
PC5	S3	fa0/2
S3	R2	fa0/24 → fa2/0

Serial DTE Connections

From	To	Port/Interface
R0	R1	se0/0 ↔ se1/0
R1	R2	se0/0 ↔ se1/0

- **OSPF Routing Table**

Copper Straight-Through Connections

From	To	Port/Interface
PC0	S1	fa0/1
PC1	S1	fa0/2
S1	R0	fa0/24 → fa2/0
PC2	S2	fa0/1
PC3	S2	fa0/2
S2	R1	fa0/24 → fa2/0
PC4	S3	fa0/1
PC5	S3	fa0/2
S3	R2	fa0/24 → fa2/0

Serial DTE Connections

From	To	Port/Interface
R0	R1	se0/0 ↔ se1/0
R1	R2	se0/0 ↔ se1/0

📘 Simulation of Network Operation

🔹 Simulation Mode in Cisco Packet Tracer

The network was tested using Simulation Mode in Cisco Packet Tracer to observe how packets travel between devices.

Steps performed:

1. Switched from Real-Time Mode → Simulation Mode
2. Selected ICMP (ping packets) from the event list
3. Sent a ping from PC0 to PC3 (192.168.2.13) and PC4 (192.168.3.14)
4. Used the "Capture/Forward" button to track packet movement step-by-step

🔹 Working of RIP (Simulation)

When using Routing Information Protocol (RIP):

1. Initial Packet Drop
   - The first ping packet is sometimes lost
   - This happens because routers are still learning routes
2. Routing Table Exchange
   - Routers exchange full routing tables periodically
   - Each router updates its table using hop count
3. Packet Flow
   - Packet travels:
     - PC0 → S1 → R0 → R1 → Destination (PC3)
   - For longer path:
     - PC0 → R0 → R1 → R2 → PC4
4. Observation
   - Slower packet delivery initially
   - More delay due to periodic updates

🔹 Working of OSPF (Simulation)

When using Open Shortest Path First (OSPF):

1. Fast Convergence
   - Routers quickly share link-state information
   - Network map is built faster
2. Shortest Path Selection
   - OSPF selects path based on cost (bandwidth)
   - Always chooses optimal route
3. Packet Flow
   - Packet directly follows the best path
   - If multiple paths exist, best one is chosen

4. Observation
    - Faster response compared to RIP
    - Less unnecessary traffic

- Step-by-Step Packet Flow (Example)

Ping: PC0 → PC4 (192.168.3.14)

Step	Device	Action
1	PC0	Sends ICMP request
2	S1	Forwards to R0
3	R0	Checks routing table
4	R1	Forwards packet
5	R2	Sends to destination network
6	PC4	Receives and replies
7	Reverse Path	Reply returns to PC0

Output:

Routing Information Protocol (RIP)

- Route Table

Copper Straight-Through Connections

From	To	Port/Interface
PC0	S1	fa0/1
PC1	S1	fa0/2
S1	R0	fa0/24 → fa2/0
PC2	S2	fa0/1
PC3	S2	fa0/2
S2	R1	fa0/24 → fa2/0
PC4	S3	fa0/1
PC5	S3	fa0/2
S3	R2	fa0/24 → fa2/0

Serial DTE Connections

From	To	Port/Interface
R0	R1	se0/0 ↔ se1/0
R1	R2	se0/0 ↔ se1/0

- Ping Result 1 (192.168.2.13)

Packet	Result	Time	TTL
1	Reply	1 ms	126
2	Reply	2 ms	126
3	Reply	2 ms	126
4	Reply	11 ms	126

Statistics

Metric	Value
Sent	4
Received	4
Lost	0 (0%)
Minimum Time	1 ms
Maximum Time	11 ms
Average Time	4 ms

- Ping Result 2 (192.168.3.14)

Packet	Result	Time	TTL
1	Timeout	—	—
2	Reply	11 ms	125

Packet	Result	Time	TTL
3	Reply	10 ms	125
4	Reply	12 ms	125

Statistics

Metric	Value
Sent	4
Received	3
Lost	1 (25%)
Minimum Time	10 ms
Maximum Time	12 ms
Average Time	11 ms

📘 Open Shortest Path First (OSPF)

- Route Table

Copper Straight-Through Connections

From	To	Port/Interface
PC0	S1	fa0/1
PC1	S1	fa0/2
S1	R0	fa0/24 → fa2/0
PC2	S2	fa0/1
PC3	S2	fa0/2
S2	R1	fa0/24 → fa2/0
PC4	S3	fa0/1
PC5	S3	fa0/2
S3	R2	fa0/24 → fa2/0

Serial DTE Connections

From	To	Port/Interface
R0	R1	se0/0 ↔ se1/0
R1	R2	se0/0 ↔ se1/0
R0	R2	se1/0 ↔ se0/0

- Ping Result 1 (192.168.2.12)

Packet	Result	Time	TTL
1	Timeout	—	—
2	Reply	1 ms	126
3	Reply	11 ms	126
4	Reply	3 ms	126

Statistics

Metric	Value
Sent	4
Received	3
Lost	1 (25%)
Minimum Time	1 ms
Maximum Time	11 ms
Average Time	5 ms

- Ping Result 2 (192.168.3.14)

Packet	Result	Time	TTL
1	Timeout	—	—
2	Reply	11 ms	125
3	Reply	32 ms	125
4	Reply	10 ms	125

Statistics

Metric	Value
Sent	4
Received	3
Lost	1 (25%)
Minimum Time	10 ms
Maximum Time	32 ms
Average Time	17 ms